

ETC5512: Finding Open Data and Assessing its Quality

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Learning Objectives

- [LO2] Practice searching for new data sources
- [LO1] Identify allowable usage and licensing of data
- [LO4] Learn about a range of different data formats
- [LO6] Recognise the components needed for effective curation of data.

Tutorial Context: Bushfires.

There is a lot of different data required to understand bush fires and their impacts on communities. This data ranges from data about the weather, to data about fire exposure, to data about how the fire will interact with peoples socio-personal contexts.

Today we are going to look at one data set that is needed in an emergency bushfire scenario.

What are neighbourhood safer places?

Neighbourhood safer places (NSPs) are a place of last resort if all other fire plans have failed and are places that may provide some protection from direct flame and heat from a fire. They do **not** guarantee safety and are **not** an alternative to planning to leave early or to stay and defend your property in the event of bushfire.

They used to be called bushfire places of last resort. You will see signs for them as you drive around Victoria.

The knowledge of where these locations are is important to communicate with people in fire impacted regions.

Looking after this data is therefore in the public interest.

In Lecture 3, we will talk more about the important role of open data for supporting effective early warning systems.

Exercise 1: Neighbourhood Safer Places (Victoria)

- i. Find a data set (ideally .csv or equivalent) that provides the locations of Neighbourhood Safer Places in Victoria?

Do this exercise as a class and search different places. Discuss what data your find and what data is most suitable.

Some places to start looking ...

- <https://www.data.vic.gov.au/>
- <https://data.gov.au/>
- Use a Search Engine

- ii. When is this data accurate to?
- iii. Is this data machine readable?
- iv. What is the licence of this data?
- v. Does this data set have meta data?
- vi. Is there a legal requirement to provide this data?

Exercise 2: Data Best Practices

FAIR Data and 5-star quality

Remember: Back in lecture 1 we learnt about FAIR data and 5-star quality data.

- i. Would you say the data in exercise 1 is:
 - Findable - Yes / No / Partly
 - Accessible - Yes / No / Partly
 - Interopable - Yes / No / Partly
 - Reusable - Yes / No / Partly
- ii. What star rating would you give this data?

Checking our understanding of AI

Can AI do this same exercise?

No - [Take a look at this prompt](#)

It tried, but it returned data that was not up-to-date! In an emergency scenario out of date data or providing people with the incorrect information could impact people's lives. It also returned a made up pdf link.

AI can help you along with your thinking, but you must always critically evaluate the answers it gives you and whether they are suitable!

It is recommended that you refer to the linked resources that the AI cites as its training data to see if these are verified sources of information.

In your own time

R practice

Work through the following [startR modules](#):

- Do the module on Projects and Paths (Module 4). *From this week onward we will assume you know how to use RProjects and why these help us organise our analytics work.*
- Do the module on Strategies for troubleshooting R (Module 5).

These should take you ~ 50 minutes.

Additional exercises

Repeat the above exercise for New South Wales or South Australia (see links at the bottom of the CFA website for Victoria). You'll find similar, but different issues with the data available.

What other data sets do you think might be needed in a emergency bushfire scenario? Is that data open?

Citations

How to cite data?

Anytime you use a data set we expect you to cite it.

We don't mind what citation style you use, provided you use it consistently across your references.

There are lots of resources available from [Monash Library](#).

Here are links for [APA7](#) and [Harvard styles](#).

Country Fire Authority Victoria. (2026). NSP-BPLR - List of Locations. [Data set]. Retrieved March 4th, 2026, from <https://www.cfa.vic.gov.au/ArticleDocuments/441/Designated%20NSP-BPLR%20-%20CFA%20Website%20complete%20list%20%2002-02-2026.pdf>

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